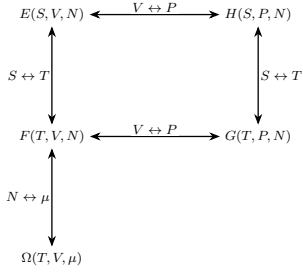


# Statistical Mechanics — Formula Sheet

Exam 77802 A, 2026

## Thermodynamics and potentials

**Thermodynamic potentials.** Legendre:  $p = \partial f / \partial x$ ,  $f^*(p) = px(p) - f(x(p))$ .



$$\begin{aligned} E &= TS - PV + \mu N & dE &= TdS - PdV + \mu dN \\ H &= E + PV = TS + \mu N & dH &= TdS + VdP + \mu dN \\ F &= E - TS = -PV + \mu N & dF &= -SdT - PdV + \mu dN \\ G &= E - TS + PV = \mu N & dG &= -SdT + VdP + \mu dN \\ \Omega &= F - \mu N = -PV & d\Omega &= -SdT - PdV - Nd\mu \end{aligned}$$

$E = TS - PV + \mu N$  is the Euler relation for an extensive homogeneous system.

**Processes:** isothermal  $dT = 0$ ; isochoric  $dV = 0$ ; isobaric  $dP = 0$ ; adiabatic  $\delta Q = 0$ ; isentropic  $dS = 0$ . For a reversible path  $dS = \delta Q_{\text{rev}}/T$ ; generally  $dS \geq \delta Q/T$ . A reversible adiabatic process is isentropic.

## Thermodynamic quantities

$$\begin{aligned} \beta &= \frac{1}{k_B T}, & \frac{\partial \beta}{\partial T} &= -\frac{1}{T^2 k_B} = -\beta^2 k_B, & C &= \frac{\partial U}{\partial T}. \\ J_i &\equiv \frac{\partial E(S, x_i)}{\partial x_i}, & dE &= T dS + \sum_i J_i dx_i. \end{aligned}$$

$$E(S, x_i) = T(S, x_i)S + \sum_i J_i x_i \quad SdT + \sum_i x_i dJ_i = 0.$$

## Entropy and maximum entropy

**Boltzmann:** for an isolated macrostate with multiplicity  $\Omega$ ,

$$S_B = k_B \ln \Omega, \quad \frac{1}{T} = \left( \frac{\partial S}{\partial E} \right)_{V, N}.$$

**Gibbs–Shannon:**

$$S[p] = -k_B \sum_i p_i \ln p_i, \quad \sum_i p_i = 1.$$

Maximizing  $S[p]$  subject to  $\langle A_a \rangle = \sum_i p_i A_{a,i}$  gives

$$p_i = \frac{1}{Z} \exp \left( - \sum_a \lambda_a A_{a,i} \right), \quad Z = \sum_i \exp \left( - \sum_a \lambda_a A_{a,i} \right).$$

Canonical constraint  $\langle E \rangle$ :  $\lambda_E = \beta = 1/(k_B T)$  and  $p_i = Z^{-1} e^{-\beta E_i}$ . Equal probabilities apply in the *isolated total system*; tracing out a heat bath produces the Boltzmann factor  $\Omega_B(E_{\text{tot}} - E_i) \propto e^{-\beta E_i}$ .

## Transforms and ensemble choice

**Legendre transform:** if  $p = f'(x)$ ,  $f^*(p) = px(p) - f(x(p))$ . It exchanges a controlled extensive variable with its intensive conjugate.

**Laplace transform:**

$$\mathcal{L}\{g\}(s) = \int_0^\infty e^{-sx} g(x) dx.$$

Canonical  $Z(\beta)$  is the Laplace transform of the density of states:  $Z(\beta) = \int \rho(E) e^{-\beta E} dE$ .

**z-transform / generating function:**

$$\tilde{a}(z) = \sum_{N \geq 0} a_N z^N, \quad Z_\Omega(z) = \sum_{N \geq 0} z^N Z_F, \quad z = e^{\beta \mu}.$$

Mostly used in the context of cluster expansion.

| fixed variables | ensemble            | potential  |
|-----------------|---------------------|------------|
| $E, V, N$       | microcanonical      | $E$ or $S$ |
| $T, V, N$       | canonical           | $F$        |
| $T, P, N$       | isothermal–isobaric | $G$        |
| $T, V, \mu$     | grand canonical     | $\Omega$   |

## Partition functions and derivatives

With weights  $z^i$  and  $Z = \sum_i z^i$ ,

$$\langle A \rangle \equiv \sum_i A_i p_i = Z^{-1} \sum_i A_i z^i.$$

**Canonical ensemble** ( $T, V, N$ )

$$Z_F = \sum_i e^{-\beta E_i}, \quad F = -k_B T \ln Z_F, \quad U = -\partial_\beta \ln Z_F,$$

$$S = k_B (\ln Z_F + \beta U), \quad P = k_B T \partial_V \ln Z_F, \\ \text{Var}(E) = \partial_\beta^2 \ln Z_F = k_B T^2 C_V.$$

**Cumulants / generating functions.**

$$M_X(t) \equiv \langle e^{tX} \rangle, \quad K_X(t) \equiv \ln M_X(t),$$

$$\langle X^n \rangle = M_X^{(n)}(0), \quad \kappa_n(X) = K_X^{(n)}(0).$$

Canonical energy:  $M_E(t) = Z_F(\beta - t)/Z_F(\beta)$ , hence

$$\kappa_n(E) = (-1)^n \partial_\beta^n \ln Z_F,$$

so  $\langle E \rangle = -\partial_\beta \ln Z_F$  and  $\text{Var}(E) = \partial_\beta^2 \ln Z_F$ . Grand canonical:

$$\kappa_n(N) = \partial_{(\beta \mu)}^n \ln Z_\Omega = (z \partial_z)^n \ln Z_\Omega.$$

For nonnegative integer  $N$ , the PGF is  $G_N(z) = \langle z^N \rangle = \sum_n P_n z^n$ ,

with  $M_N(t) = \langle e^{tN} \rangle = G_N(e^t)$ ,  $P_n = G_N^{(n)}(0)/n!$ ,  $G'_N(1) = \langle N \rangle$ ,

$G''_N(1) = \langle N(N-1) \rangle$ .

Classical phase space:

$$Z_F = \frac{1}{N! h^{3N}} \int \prod_{i=1}^N d^3 p_i d^3 q_i e^{-\beta H_N}.$$

**Gibbs ensemble** ( $T, P, N$ )

$$Z_G(\beta, P, N) = \int_0^\infty dV e^{-\beta PV} Z_F(\beta, V), \quad G = -k_B T \ln Z_G.$$

Here the effective energy is the enthalpy (Laplace transform of  $Z_F$  in  $V$ )

$$H = E + PV \\ \Rightarrow \langle H \rangle = -\partial_\beta \ln Z_G, \quad \langle V \rangle = -\frac{1}{\beta} \partial_P \ln Z_G,$$

$$S = k_B \ln Z_G + k_B \beta \langle H \rangle.$$

For spins in a magnetic field,

$$B \leftrightarrow -P, \quad M = \sum_i s_i \leftrightarrow V, \quad -BM \leftrightarrow PV.$$

**Grand-canonical ensemble** ( $T, V, \mu$ )

$$Z_\Omega(\beta, V, \mu) = \sum_{N=0}^\infty e^{\beta \mu N} Z_F(\beta, V, N) = \sum_N z^N Z_F(\beta, V, N)$$

The grand potential is

$$\Omega = -\frac{1}{\beta} \ln Z_\Omega = -PV$$

The mean particle number and its fluctuations are

$$\langle N \rangle = \frac{1}{\beta} \partial_\mu \ln Z_\Omega = z \partial_z \ln Z_\Omega,$$

$$\text{Var}(N) = \frac{1}{\beta^2} \partial_\mu^2 \ln Z_\Omega.$$

The entropy is

$$S = k_B \ln Z_\Omega + k_B \beta (\langle E \rangle - \mu \langle N \rangle).$$

## Ideal gas and equipartition

Thermal wavelength  $\lambda_T = h/\sqrt{2\pi m k_B T}$ :

$$Z_F = \frac{1}{N!} \left( \frac{V}{\lambda_T^3} \right)^N, \quad PV = N k_B T, \quad U = \frac{3}{2} N k_B T.$$

Using Stirling,  $\ln N! \simeq N \ln N - N$ :

$$F = -N k_B T \left[ \ln \left( \frac{V}{N \lambda_T^3} \right) + 1 \right], \quad S = N k_B \left[ \ln \left( \frac{V}{N \lambda_T^3} \right) + \frac{5}{2} \right].$$

Every independent quadratic term in  $H$  contributes  $\frac{1}{2} k_B T$  to  $U$  and  $\frac{1}{2} k_B$  to  $C_V$ . More generally (boundary terms vanishing),

$$\langle x_i \partial_{x_j} H \rangle = k_B T \delta_{ij},$$

so  $H \supset a x^2$  gives  $a \langle x^2 \rangle = \frac{1}{2} k_B T$  (or  $H \supset \frac{1}{2} a x^2$  gives  $\frac{1}{2} a \langle x^2 \rangle = \frac{1}{2} k_B T$ ).

## Laplace / saddle-point method

For  $N \gg 1$ ,  $e^{-Nf(x)}$  is dominated by the minimum  $x_*$  of  $f$ :

$$I(N) = \int g(x) e^{-Nf(x)} dx \simeq g(x_*) e^{-Nf(x_*)} \sqrt{\frac{2\pi}{N f''(x_*)}}.$$

First correction ( $f'(x_*) = 0$ , write  $f_* = f(x_*)$ , etc.):

$$I(N) \sim e^{-Nf_*} \sqrt{\frac{2\pi}{N f''_*}} \left[ g_* + \frac{1}{N} \delta_1 + \dots \right],$$

$$\delta_1 = \frac{g''_*}{2f''_*} - \frac{g'_* f'''_*}{2(f''_*)^2} - \frac{g_* f''''_*}{8(f''_*)^2} + \frac{5g_* (f''_*)^2}{24(f''_*)^3}.$$

In  $d$  dimensions, with Hessian  $H_{ij} = \partial_i \partial_j f(x_*)$ ,

$$I(N) \simeq g(x_*) e^{-Nf(x_*)} \left( \frac{2\pi}{N} \right)^{d/2} (\det H)^{-1/2}.$$

Width  $\Delta x \sim [N f''(x_*)]^{-1/2}$ ; **small parameter:**  $1/N$  (or  $\beta$  at low  $T$ ). Boundary / degenerate minima need a modified expansion.

$$H = -J \sum_{i=1}^N s_i s_{i+1} - h \sum_{i=1}^N s_i, \quad s_i = \pm 1, \quad s_{N+1} = s_1.$$

Split the field equally between adjacent bonds:

$$T_{ss'} = \exp \left[ \beta J s s' + \frac{\beta h}{2} (s + s') \right], \quad T = \begin{pmatrix} e^{\beta J + \beta h} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J - \beta h} \end{pmatrix}.$$

$$Z_N = \text{Tr } T^N = \lambda_+^N + \lambda_-^N,$$

$$\lambda_{\pm} = e^{\beta J} \cosh(\beta h) \pm \sqrt{e^{2\beta J} \sinh^2(\beta h) + e^{-2\beta J}}.$$

For  $N \rightarrow \infty$ ,  $f = -k_B T \ln \lambda_+$  and  $m = \partial_{\beta h} \ln \lambda_+$ . Insert the diagonal spin operator  $S = \text{diag}(1, -1)$ :

$$\langle s_0 s_r \rangle = \frac{\text{Tr}(S T^r S T^{N-r})}{\text{Tr } T^N}.$$

If  $T = \sum_a \lambda_a |a\rangle \langle a|$  with  $|\lambda_0| > |\lambda_1| \geq \dots$ , then for  $N \rightarrow \infty$

$$\langle A_0 B_r \rangle = \sum_a \left( \frac{\lambda_a}{\lambda_0} \right)^r \langle 0 | A | a \rangle \langle a | B | 0 \rangle,$$

so  $\langle A_0 B_r \rangle_c \sim (\lambda_1/\lambda_0)^r$  and  $\xi^{-1} = -\ln |\lambda_1/\lambda_0|$ . **At  $h = 0$ :** with  $q \equiv \lambda_-/\lambda_+$ ,

$$\langle s_0 s_r \rangle = \frac{q^r + q^{N-r}}{1 + q^N}.$$

As  $N \rightarrow \infty$ :

$$\langle s_0 s_r \rangle = q^r = (\tanh \beta J)^r = e^{-r/\xi}, \quad \xi^{-1} = -\ln |q| = -\ln(\tanh \beta J).$$

A phase transition would require correlations to persist over arbitrarily large distances, i.e.  $\xi \rightarrow \infty$ . In the 1D Ising model,  $|\lambda_1| < \lambda_0$  for every  $T > 0$ , so  $\xi$  remains finite and there is no finite-temperature transition. Generally, the subleading eigenvalue sets the slowest-decaying correction and therefore controls long-range correlations.

### Ising mean-field theory

Consider a system with nearest-neighbor interactions,

$$H_{\text{int}} = -J \sum_{\langle i,j \rangle} s_i s_j.$$

Define the fluctuation around the mean magnetization by

$$\delta s_i = s_i - m.$$

Let  $d$  be the spatial dimension and  $z = 2d$  the coordination number, i.e. the number of nearest neighbors of each site.

**Partition function:**

$$Z_G = \text{tr} \exp \left\{ \beta \left( J \sum_{\langle i,j \rangle} s_i s_j + B \sum_i s_i \right) \right\}.$$

Using  $s_i = m + \delta s_i$ , using  $[\dots] = \delta s_i \delta s_j + m(\delta s_i + \delta s_j) + m^2$

$$Z_G = \text{tr} \exp \left\{ \beta J \sum_{\langle i,j \rangle} [\dots] + \beta B \sum_i \delta s_i + \beta B N m \right\}.$$

**Mean-field assumption:**

$$\delta s_i \delta s_j \approx 0 \implies s_i s_j \approx m(s_i + s_j) - m^2.$$

**Approximate partition function:** Since

$$\sum_{\langle i,j \rangle} 1 = \frac{Nz}{2},$$

we obtain

$$Z_G^{\text{MF}} = \exp \left( -\frac{\beta J N z m^2}{2} \right) [2 \cosh(\beta B_{\text{MF}})]^N,$$

where

$$B_{\text{MF}} = B + J z m.$$

The self-consistency equation for the magnetization is

$$m = \frac{1}{\beta N} \frac{\partial \ln Z_G}{\partial B} = \tanh[\beta(B + J z m)].$$

The mean-field free energy per spin (extrema give that equation) is

$$f_{\text{MF}}(m) = \frac{Jz}{2} m^2 - \frac{1}{\beta} \ln[2 \cosh(\beta(B + J z m))]$$

(up to an additive constant). Then

$\partial_m f_{\text{MF}} = 0 \Leftrightarrow m = \tanh[\beta(B + J z m)]$ , and  $\partial_m^2 f_{\text{MF}} > 0$  selects stable solutions.

**Power-laws** At  $h = 0$ ,  $T_c = zJ/k_B$ . Near  $T_c$ :

$$m \sim (-t)^{1/2}, \quad \chi \sim |t|^{-1}, \quad m(T_c, h) \sim h^{1/3}, \quad t = (T - T_c)/T_c.$$

### Hubbard–Stratonovich transformation

For  $a > 0$ ,

$$e^{\frac{a}{2} x^2} = \frac{1}{\sqrt{2\pi a}} \int_{-\infty}^{\infty} d\phi \exp \left[ -\frac{\phi^2}{2a} + \phi x \right].$$

For a positive matrix  $K$ ,

$$e^{\frac{1}{2} \mathbf{s}^T K \mathbf{s}} = \frac{1}{\sqrt{\det(2\pi K)}} \int d^n \phi e^{-\frac{1}{2} \phi^T K^{-1} \phi + \phi^T \mathbf{s}}.$$

The Hubbard–Stratonovich transformation trades a direct quadratic interaction between the microscopic variables for a linear interaction with a new auxiliary field. The new field  $\phi$  is continuous and is integrated over all real values.

After the microscopic variables are summed or integrated out,  $\phi$  acquires an effective action and describes fluctuations of the order parameter. Approximating the remaining integral by its dominant saddle point fixes  $\phi$  to its most probable value and reproduces the mean-field approximation.

### Landau and Ginzburg–Landau theory

Landau theory uses a coarse-grained order parameter  $\varphi$  (e.g. magnetization). (After HS, integrating out the microscopic degrees of freedom turns the auxiliary field  $\phi$  into this Landau order parameter  $\varphi$ .) Above  $T_c$ , symmetry favors  $\varphi = 0$ ; below  $T_c$  the system may choose  $\varphi \neq 0$ . Expand the free-energy density about  $\varphi = 0$ :

$$f(T, \varphi) = f_0(T) + \frac{1}{2}a(T)\varphi^2 + \frac{1}{4}u\varphi^4 - h\varphi + \dots,$$

with  $u > 0$  and  $a(T) \approx a_0(T - T_c) = a_0 T_c t$ ,  $t = (T - T_c)/T_c$ ,  $a_0 > 0$ . Equivalently,

$$f(\varphi) = f_0 + \frac{a_0 t}{2}\varphi^2 + \frac{u}{4}\varphi^4 - h\varphi + \dots.$$

**Minimize:** at  $h = 0$ ,

$$\frac{\partial f}{\partial \varphi} = 0 \implies a(T)\varphi + u\varphi^3 = 0.$$

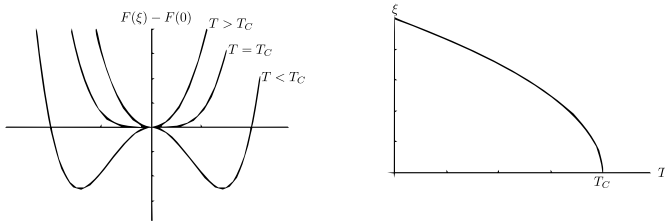
Solutions:  $\varphi = 0$  always, and  $\varphi = \pm\sqrt{-a(T)/u}$  when  $a(T) < 0$  (i.e.  $T < T_c$ ). Near  $T_c$ ,

$$\varphi(T) = \pm\sqrt{\frac{a_0(T_c - T)}{u}} \sim (-t)^{1/2}, \quad \Rightarrow \quad \beta_{\text{MF}} = \frac{1}{2}.$$

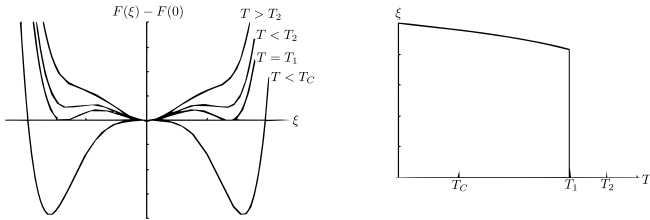
If  $u > 0$  the order parameter grows continuously from zero (**second-order** / continuous transition). If the quartic coefficient is negative, a stabilizing  $+v\varphi^6$  term is needed and the transition is first order;  $u = 0$  is the tricritical point.

**Phase classification** (order parameter  $\xi$ ):

Second Order Phase Transition



First Order Phase Transition



Ginzburg–Landau allows  $\varphi$  to vary in space:

$$\mathcal{F}[\varphi] = \int d^d x \left[ \frac{c}{2}(\nabla\varphi)^2 + \frac{r}{2}\varphi^2 + \frac{u}{4}\varphi^4 - h\varphi \right], \quad Z = \int \mathcal{D}\varphi e^{-\beta\mathcal{F}[\varphi]}.$$

The gradient penalizes rapid spatial changes; the remaining terms set the locally preferred  $\varphi$ . Minimizing  $\mathcal{F}$  gives

$$-c\nabla^2\varphi + r\varphi + u\varphi^3 = h.$$

Valid near a continuous transition for slow variations on scales  $\gg$  lattice spacing (long-wavelength order-parameter fluctuations).

### Critical exponents, scaling and universality

Near a continuous transition, physical quantities follow power laws:

$$C \sim |t|^{-\alpha}, \quad m \sim (-t)^\beta, \quad \chi \sim |t|^{-\gamma}, \quad m(t=0, h) \sim h^{1/\delta}, \\ \xi \sim |t|^{-\nu}, \quad G(r, t=0) \sim r^{-(d-2+\eta)}.$$

These powers define the critical exponents. They appear because, as  $t \rightarrow 0$ , the correlation length diverges and the system becomes approximately scale invariant. Widom's scaling hypothesis assumes that the singular part of the free-energy density has the form

$$f_s(t, h) = |t|^{2-\alpha} \mathcal{F}_\pm(h/|t|^\delta).$$

Thus  $t$  and  $h$  do not enter independently, but only through a scale-free combination. Taking derivatives of  $f_s$ , for example

$$m = -\partial_h f_s, \quad \chi = \partial_h m,$$

produces the power laws and relations between their exponents:

$$m = |t|^\beta \mathcal{M}_\pm(h/|t|^\delta),$$

$\alpha + 2\beta + \gamma = 2$ ,  $\gamma = \beta(\delta - 1)$ ,  $\gamma = \nu(2 - \eta)$ ,  $2 - \alpha = d\nu$ . Mean-field values for scalar  $\phi^4$  theory are (see 3 boxes above)

$$(\alpha, \beta, \gamma, \delta, \nu, \eta) = (0, \frac{1}{2}, 1, 3, \frac{1}{2}, 0).$$

Universality means that the exponents depend mainly on spatial dimension, symmetry, number of order-parameter components and interaction range, rather than on microscopic details.

### Fourier-space intuition

For a lattice field  $s_i$  at positions  $\mathbf{r}_i$ ,

$$s_i = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} s_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_i}, \quad s_{\mathbf{k}} = \frac{1}{\sqrt{N}} \sum_i s_i e^{-i\mathbf{k} \cdot \mathbf{r}_i}.$$

Translation-invariant quadratic forms diagonalize:

$$\frac{1}{2} \sum_{ij} s_i K_{i-j} s_j = \frac{1}{2} \sum_{\mathbf{k}} K(\mathbf{k}) |s_{\mathbf{k}}|^2.$$

Near a transition the soft modes have  $ka \ll 1$ , so

$K(\mathbf{k}) \simeq r + ck^2 + \dots$ ; inverse Fourier transformation gives the Ginzburg–Landau gradient term.

## Correlations, Goldstone Modes and Interacting Gases

### Gaussian functional integrals and Green functions

For a positive operator  $K$  and dimensionless source  $J$ ,

$$Z[J] = \int \mathcal{D}\phi \exp \left[ -\frac{1}{2} \phi K \phi + J\phi \right] = Z[0] \exp \left( \frac{1}{2} J K^{-1} J \right).$$

$$\langle \phi(x) \rangle_J = \frac{\delta \ln Z}{\delta J(x)},$$

$$G_c(x, y) = \langle \phi(x) \phi(y) \rangle_J = \frac{\delta^2 \ln Z}{\delta J(x) \delta J(y)} = K^{-1}(x, y).$$

For the Gaussian Ginzburg–Landau theory,

$$K(\mathbf{k}) = r + ck^2, \quad \tilde{G}(\mathbf{k}) = \frac{1}{r + ck^2}, \quad \xi = \sqrt{c/r}.$$

At large  $R = |\mathbf{x}|$ ,

$$G(R) \sim \frac{e^{-R/\xi}}{R^{(d-1)/2}} \quad (R \gg \xi), \quad G(R) \sim R^{-(d-2)} \quad (r = 0, d > 2).$$

If the physical field couples as  $-h \int \phi$ , then

$$\chi = \frac{\partial \langle \phi \rangle}{\partial h} = \beta \int d^d x G_c(x, 0) = \beta \tilde{G}(0).$$

When  $J = \beta h$  is used as the source, the same statement is

$$\partial \langle \phi \rangle / \partial J = \tilde{G}(0).$$

### Goldstone modes

An  $O(n)$  field is an  $n$ -component real field,  $\vec{\phi}(\vec{x}) = (\phi_1(\vec{x}), \dots, \phi_n(\vec{x}))$ , whose free energy is invariant under rotations  $R \in O(n)$ .

For an  $O(n)$  field,

$$\mathcal{F} = \int d^d x \left[ \frac{c}{2}(\nabla\phi)^2 + \frac{r}{2}\phi^2 + \frac{u}{4}(\phi^2)^2 \right], \quad r < 0.$$

The minima form a continuous sphere of equivalent ordered states. Spontaneous symmetry breaking means that the system chooses one direction, for example

$$\langle \phi \rangle = v \hat{e}_1, \quad v^2 = -r/u,$$

and we decompose

$$\phi = (v + \sigma, \pi_1, \dots, \pi_{n-1}).$$

The longitudinal fluctuation  $\sigma$  changes the magnitude of the order parameter and moves away from the minimum, so it costs local energy. The transverse fluctuations  $\pi_a$  rotate the order parameter between equivalent minima, so a uniform rotation costs no energy:

$$m_L^2 = 2|r| \quad (\sigma, \text{longitudinal}), \quad m_T^2 = 0 \quad (\pi_a, \text{transverse}).$$

Thus spontaneous breaking  $G \rightarrow H$  creates  $\dim G - \dim H$  massless Goldstone modes: one for each independent direction along the degenerate vacuum manifold. For  $O(n) \rightarrow O(n-1)$ , there are  $n-1$  such modes, with

$$G_T(k) \sim \frac{1}{ck^2}.$$

Only spatial variations of these rotations cost gradient energy, which vanishes for sufficiently long wavelengths.

Mermin–Wagner theorem: In local-interaction systems, there is no finite-temperature continuous symmetry breaking in  $d < 2$  dimensions.

### Upper/lower critical dimensions and Ginzburg criterion

The lower critical dimension asks: in how few dimensions can an ordered phase survive fluctuations? For a system with a broken continuous symmetry, the massless Goldstone modes have

$$G_T(k) \sim \frac{1}{k^2}, \quad \langle \pi^2 \rangle \sim \int_{k_{\min}} \frac{d^d k}{k^2}.$$

For  $d \leq 2$ , the long-wavelength contribution  $k \rightarrow 0$  diverges.

Large-scale rotations then destroy the chosen direction of the order parameter, so there is no finite- $T$  continuous symmetry breaking in short-range systems (Mermin–Wagner). Hence  $d_\ell = 2$  for  $O(n \geq 2)$ . For the discrete Ising symmetry there are no Goldstone modes, and  $d_\ell = 1$ .

The upper critical dimension asks: above which dimension are fluctuations weak enough that mean-field theory becomes correct? For  $\phi^4$ ,

$$g(\xi) \sim \frac{uk_B T}{c^2} \xi^{4-d}$$

measures the strength of fluctuations on the correlation-length scale. Mean field is self-consistent when  $g(\xi) \ll 1$  (Ginzburg criterion). Since

$\xi \rightarrow \infty$  near the transition, fluctuations grow for  $d < 4$ , are marginal at  $d = 4$ , and become irrelevant for  $d > 4$ . Therefore  $d_c = 4$ : below it the exponents are non-mean-field, at it logarithmic corrections appear, and above it mean-field theory is asymptotically correct. Hyperscaling generally fails above  $d_c$ .

#### Mayer expansion and linked clusters

For pair interactions  $U = \sum_{i < j} u(r_{ij})$  define

$$f_{ij} = e^{-\beta u(r_{ij})} - 1, \quad e^{-\beta U} = \prod_{i < j} (1 + f_{ij}).$$

Each term corresponds to a graph: particles are vertices and  $f_{ij}$  are edges. The configurational integral is a sum over all graphs. Every graph factorizes into connected components; exponentiating the connected-cluster sum gives

$$\ln Z_\Omega = V \sum_{\ell \geq 1} b_\ell z^\ell, \quad \beta P = \sum_{\ell \geq 1} b_\ell z^\ell, \quad n = \sum_{\ell \geq 1} \ell b_\ell z^\ell,$$

where this convention absorbs the powers of  $\lambda_T$  into  $z, b_\ell$ . The cluster coefficients keep only *connected* graphs  $\mathcal{C}_\ell$ :

$$b_\ell = \frac{1}{\ell! V} \int_{V^\ell} \left( \sum_{G \in \mathcal{C}_\ell} \prod_{(i,j) \in G} f_{ij} \right) \prod_{i=1}^{\ell} d^d r_i.$$

The logarithm drops disconnected graphs because they factorize into products already generated by the exponential.

**Example**  $\ell = 3$ : The connected graphs are the three open chains and the triangle:

$$\sum_{G \in \mathcal{C}_3} \prod f_{ij} = f_{12}f_{13} + f_{12}f_{23} + f_{13}f_{23} + f_{12}f_{13}f_{23}.$$

Hence

$$b_3 = \frac{1}{3!V} \int_{V^3} [f_{12}f_{13} + f_{12}f_{23} + f_{13}f_{23} + f_{12}f_{13}f_{23}] \prod_{i=1}^3 d^d r_i.$$

The open chains are reducible: removing their central particle disconnects the graph. The triangle remains connected after removing any one particle and is therefore irreducible. Connected graphs determine  $\ln Z_\Omega$ , while the virial coefficients contain only irreducible graphs; consequently, only the triangle contributes directly to  $B_3$ . For a partition with  $n_k$  clusters of size  $k$ ,

$$\sum_k k n_k = N, \quad \Omega_N(\{n_k\}) = \frac{N!}{\prod_k (k!)^{n_k} n_k!}.$$

If unlabeled particles, must take into account topology - multiply  $b_\ell$  by  $\ell! / |\text{Aut}(G)|$ .

#### Virial expansion and basic coefficients

The grand-canonical calculation naturally gives pressure and density as functions of the fugacity  $z$ . Eliminating  $z$  between  $P(z)$  and  $n(z)$  gives the virial expansion,

$$\beta P = n + B_2 n^2 + B_3 n^3 + \dots$$

The first term is the ideal-gas law; the higher powers describe corrections due to interactions between pairs, triples, and larger groups of particles. If

$$\beta P = z + b_2 z^2 + b_3 z^3 + \dots,$$

then

$$B_2 = -b_2, \quad B_3 = 4b_2^2 - 2b_3.$$

For a translation-invariant pair potential,

$$B_2 = -\frac{1}{2} \int d^d r f(r),$$

$$B_3 = -\frac{1}{3} \int d^d r_{12} d^d r_{13} f_{12} f_{13} f_{23}.$$

Thus  $B_2$  measures the effect of one interacting pair, while  $B_3$  contains the genuinely three-particle contribution in which all three particles are linked. The virial expansion is useful experimentally because density is measurable, whereas fugacity is not.

For hard spheres of diameter  $\sigma$  in three dimensions,

$$B_2 = \frac{2\pi}{3} \sigma^3, \quad B_3 = \frac{5\pi^2}{18} \sigma^6 = \frac{5}{8} B_2^2.$$

Here  $B_2$  is half the volume excluded to the center of a second sphere. More generally, for a  $d$ -dimensional hard core,

$$B_2 = \frac{1}{2} V_d(\sigma), \quad V_d(R) = \frac{\pi^{d/2} R^d}{\Gamma(d/2 + 1)}.$$

## Stochastic Processes, Noise and Path Integrals

#### Markov processes and Chapman–Kolmogorov

A process is Markovian when the future conditioned on the present is independent of the past:

$$P(x_n, t_n | x_{n-1}, t_{n-1}; \dots; x_0, t_0) = P(x_n, t_n | x_{n-1}, t_{n-1}).$$

Transition kernels obey Chapman–Kolmogorov:

$$P(x_2, t_2 | x_0, t_0) = \int dx_1 P(x_2, t_2 | x_1, t_1) P(x_1, t_1 | x_0, t_0).$$

For a homogeneous discrete chain,  $P_{n+1} = T P_n$ ,  $\sum_i T_{ij} = 1$ , where  $T_{ij}$  is the transition probability **from state  $j$  to state  $i$**  ( $j \rightarrow i$ ). (Column-stochastic: columns sum to 1.)

A stationary distribution is an eigenvector with eigenvalue 1; modes with  $|\lambda| < 1$  decay. Brownian motion is approximately Markovian when molecular collision times are much shorter than the observed time scale.

#### Wiener process and Gaussian white noise

A Wiener process  $W(t)$  has  $W(0) = 0$ , continuous paths, independent stationary increments, and

$$W(t + \Delta t) - W(t) \sim \mathcal{N}(0, \Delta t), \quad \langle W(t) \rangle = 0, \\ \langle W(t) W(s) \rangle = \min(t, s).$$

Gaussian white noise is the distributional derivative  $\eta(t) = \dot{W}(t)$ :

$$\langle \eta(t) \rangle = 0, \quad \langle \eta(t) \eta(t') \rangle = \delta(t - t').$$

For strength  $\Lambda$ , write  $\xi(t) = \sqrt{\Lambda} \eta(t)$ , so  $\langle \xi(t) \xi(t') \rangle = \Lambda \delta(t - t')$ . On a grid,

$$\langle \xi_i \xi_j \rangle = \frac{\Lambda}{\Delta t} \delta_{ij}, \quad \mathcal{P}[\xi] \propto \exp \left[ -\frac{1}{2\Lambda} \int \xi(t)^2 dt \right].$$

#### Multivariate Gaussian identities

For symmetric positive  $A$  and mean zero,

$$p(\mathbf{x}) = \frac{\sqrt{\det A}}{(2\pi)^{n/2}} \exp \left( -\frac{1}{2} \mathbf{x}^T A \mathbf{x} \right), \quad C = A^{-1}.$$

$$I(\mathbf{b}) = \langle e^{\mathbf{b}^T \mathbf{X}} \rangle = \exp \left( \frac{1}{2} \mathbf{b}^T C \mathbf{b} \right), \quad \langle X_i X_j \rangle = C_{ij}.$$

All odd centered moments vanish. Wick/Isserlis theorem:

$$\langle X_1 \cdots X_{2n} \rangle = \sum_{\pi \in \mathcal{P}_{2n}} \prod_{(i,j) \in \pi} C_{ij}, \quad \langle X_1 \cdots X_{2n+1} \rangle = 0,$$

where  $\mathcal{P}_{2n}$  is the set of perfect matchings of  $\{1, \dots, 2n\}$  (there are  $(2n-1)!! = (2n)!/(2^n n!)$  of them). E.g.

$$\langle X_1 X_2 X_3 X_4 \rangle = C_{12} C_{34} + C_{13} C_{24} + C_{14} C_{23}.$$

For Gaussian variables, zero covariance implies independence.

#### Langevin equations and moments

Underdamped Brownian particle:

$$m \dot{v} = -\gamma v + F(x) + \xi(t), \quad \langle \xi(t) \xi(t') \rangle = 2\gamma k_B T \delta(t - t').$$

For  $F = 0$  (Ornstein–Uhlenbeck),  $\tau_B = m/\gamma$  and

$$v(t) = v_0 e^{-t/\tau_B} + \frac{1}{m} \int_0^t e^{-(t-s)/\tau_B} \xi(s) ds,$$

$$\langle v \rangle = v_0 e^{-t/\tau_B}, \quad \text{Var}(v) = \frac{k_B T}{m} (1 - e^{-2t/\tau_B}).$$

Long times:  $\langle [x(t) - x(0)]^2 \rangle \simeq 2Dt$  in one dimension,  $D = k_B T/\gamma$ . Overdamped form:

$$dx = a(x, t) dt + b(x, t) dW_t.$$

For Itô calculus,

$$\frac{d}{dt} \langle A(x) \rangle = \left\langle a A' + \frac{1}{2} b^2 A'' \right\rangle.$$

Taking  $A = x, x^2, \dots$  gives moment equations without solving the full PDF.

#### Gaussian path integrals

A path integral is an ordinary multidimensional integral with one variable for each time or spatial point:

$$\int \mathcal{D}\phi \sim \lim_{N \rightarrow \infty} \int \prod_{i=1}^N d\phi_i.$$

For an ordinary Gaussian integral,

$$\int d^n x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x} + \mathbf{J}^T \mathbf{x}} = \sqrt{\frac{(2\pi)^n}{\det A}} e^{\frac{1}{2} \mathbf{J}^T A^{-1} \mathbf{J}}.$$

The functional analogue is

$$\int \mathcal{D}\phi e^{-\frac{1}{2} \int \phi K \phi + \int J \phi} \propto (\det K)^{-1/2} e^{\frac{1}{2} \int J K^{-1} J}.$$

Thus  $K^{-1}$  is the covariance, or Green's function, of the Gaussian field. For the Langevin equation

$$\dot{x}(t) = a(x, t) + \xi(t), \quad \langle \xi(t) \xi(t') \rangle = \Lambda \delta(t - t'),$$

a trajectory  $x(t)$  requires the noise realization

$$\xi(t) = \dot{x}(t) - a(x, t).$$

Since Gaussian white noise has probability

$$\mathcal{P}[\xi] \propto \exp \left[ -\frac{1}{2\Lambda} \int \xi(t)^2 dt \right],$$

the corresponding trajectory probability is

$$\mathcal{P}[x] \propto \mathcal{J}[x] \exp \left[ -\frac{1}{2\Lambda} \int (\dot{x}(t) - a(x, t))^2 dt \right].$$

Trajectories close to the deterministic equation

$$\dot{x}(t) = a(x, t)$$

are therefore the most probable.

A functional delta enforces an equation at every time:

$$\delta[F] \propto \int \mathcal{D}\hat{x} \exp \left[ i \int \hat{x}(t) F(t) dt \right].$$

For example, choosing

$$F(t) = \dot{x}(t) - a(x, t) - \xi(t)$$

imposes the Langevin equation inside the path integral.

The Jacobian  $\mathcal{J}[x]$  depends on the discretization convention and is often constant for additive or linear noise.

#### Kramers–Moyal expansion

Suppose that during a short time interval  $\tau$ , a stochastic variable changes by

$$\Delta x = x(t + \tau) - x(t),$$

given that it started at  $x(t) = x$ .

The short-time motion can be characterized by its moments:

$$\langle \Delta x \mid x(t) = x \rangle, \quad \langle (\Delta x)^2 \mid x(t) = x \rangle, \quad \langle (\Delta x)^3 \mid x(t) = x \rangle, \dots$$

The Kramers–Moyal coefficients measure how quickly these moments grow:

$$D^{(n)}(x, t) = \lim_{\tau \rightarrow 0} \frac{1}{n! \tau} \langle (\Delta x)^n \mid x(t) = x \rangle.$$

In particular,

$$D^{(1)} = \lim_{\tau \rightarrow 0} \frac{\langle \Delta x \rangle}{\tau}$$

is the local average velocity, or drift, while

$$D^{(2)} = \lim_{\tau \rightarrow 0} \frac{\langle (\Delta x)^2 \rangle}{2\tau}$$

measures the local spreading due to randomness.

These short-time moments determine the evolution of the probability density:

$$\partial_t P(x, t) = \sum_{n=1}^{\infty} (-\partial_x)^n \left[ D^{(n)}(x, t) P(x, t) \right].$$

If  $D^{(n)} = 0$  for  $n \geq 3$ , this truncates to the **Fokker–Planck equation**

$$\partial_t P = -\partial_x (D^{(1)} P) + \partial_x^2 (D^{(2)} P).$$

For the Itô stochastic equation

$$dx = a(x, t) dt + b(x, t) dW,$$

the short-time change is

$$\Delta x \simeq a\tau + b\Delta W,$$

with

$$\langle \Delta W \rangle = 0, \quad \langle (\Delta W)^2 \rangle = \tau.$$

Therefore,

$$\langle \Delta x \rangle \simeq a\tau, \quad \langle (\Delta x)^2 \rangle \simeq b^2 \tau,$$

and all higher moments vanish after division by  $\tau$  in the limit  $\tau \rightarrow 0$ . Hence,

$$D^{(1)} = a, \quad D^{(2)} = \frac{1}{2} b^2, \quad D^{(n)} = 0 \quad \text{for } n \geq 3.$$

Substituting these coefficients recovers

$$\partial_t P = -\partial_x (aP) + \frac{1}{2} \partial_x^2 (b^2 P).$$

Thus,  $a$  transports probability, while  $b$  causes probability to spread.

## Fokker–Planck, Master Equations and Monte Carlo

#### Fokker–Planck equation and probability current

For Itô  $dx = a(x, t) dt + b(x, t) dW_t$ ,

$$\partial_t P = -\partial_x [aP] + \frac{1}{2} \partial_x^2 [b^2 P] = -\partial_x J,$$

$$J = aP - \frac{1}{2} \partial_x (b^2 P).$$

In  $d$  dimensions, with  $dx_i = a_i dt + B_{i\alpha} dW_\alpha$  and  $D_{ij} = (BB^T)_{ij}$ ,

$$\partial_t P = -\partial_i (a_i P) + \frac{1}{2} \partial_i \partial_j (D_{ij} P).$$

Observable evolution (boundary terms vanishing):

$$\frac{d}{dt} \langle A(\mathbf{x}) \rangle = \langle a_i \partial_i A + \frac{1}{2} D_{ij} \partial_i \partial_j A \rangle.$$

In 1D:  $\frac{d}{dt} \langle x \rangle = \langle a(x, t) \rangle$  and  $\frac{d}{dt} \langle x^2 \rangle = 2 \langle xa \rangle + \langle b^2 \rangle$ . Boundary conditions: reflecting  $J \cdot \hat{n} = 0$ ; absorbing  $P = 0$ . Stationarity requires  $\partial_x J = 0$ ; equilibrium with no net current has  $J = 0$ . For additive overdamped motion

$$dx = \frac{F(x)}{\gamma} dt + \sqrt{2D} dW, \quad D = \frac{k_B T}{\gamma},$$

$F = -U'$  gives

$$P_{\text{eq}}(x) = Z^{-1} e^{-\beta U(x)}.$$

#### Ornstein–Uhlenbeck process

Free underdamped velocity ( $F = 0$ ),  $\tau = m/\gamma$ :

$$dv = -\frac{v}{\tau} dt + \sqrt{2D_v} dW_t, \quad v(t) = v_0 e^{-t/\tau} + \sqrt{2D_v} \int_0^t e^{-(t-s)/\tau} dW_s, \\ \langle v(t) \rangle = v_0 e^{-t/\tau}, \quad \text{Var}[v(t)] = D_v \tau (1 - e^{-2t/\tau}).$$

Thermal:  $D_v = \gamma k_B T / m^2 = k_B T / (m\tau)$ , so

$$\langle v(t)v(t') \rangle_{\text{eq}} = \frac{k_B T}{m} e^{-|t-t'|/\tau}.$$

With  $x(t) - x(0) = \int_0^t v(s) ds$  and  $D = k_B T / \gamma$ ,

$$\langle [x(t) - x(0)]^2 \rangle = 2D [t - \tau(1 - e^{-t/\tau})],$$

so  $\langle \Delta x^2 \rangle \simeq (k_B T / m) t^2$  for  $t \ll \tau$  and  $\simeq 2Dt$  for  $t \gg \tau$ .

#### Klein–Kramers equation

For

$$\dot{x} = v, \quad m\dot{v} = -\gamma v - U'(x) + \xi(t), \quad \langle \xi(t)\xi(t') \rangle = 2\gamma k_B T \delta(t - t'),$$

$$\partial_t P = -v \partial_x P + \frac{1}{m} U'(x) \partial_v P + \frac{\gamma}{m} \partial_v (vP) + \frac{\gamma k_B T}{m^2} \partial_v^2 P.$$

The stationary solution is Maxwell–Boltzmann:

$$P_{\text{eq}}(x, v) = Z^{-1} \exp \left[ -\beta \left( \frac{1}{2} m v^2 + U(x) \right) \right].$$

In the strongly overdamped limit, eliminate  $v$  to obtain the Smoluchowski Fokker–Planck equation for  $x$ .

#### Continuous-time master equation

Let  $W_{ij}$  be the transition rate  $j \rightarrow i$ . Then

$$\dot{P}_i(t) = \sum_{j \neq i} [W_{ij} P_j - W_{ji} P_i] = \sum_j L_{ij} P_j,$$

where  $L_{ij} = W_{ij}$  for  $i \neq j$  and  $L_{ii} = -\sum_{k \neq i} W_{ki}$ , so  $\sum_i L_{ij} = 0$ . A stationary state satisfies  $L\mathbf{P}^{\text{st}} = 0$ . The zero eigenvalue is the stationary mode; the nonzero eigenvalues determine relaxation times. Multiple closed communicating classes may give multiple stationary states. Probability conservation follows by summing the master equation over  $i$ .

#### Probability-generating functions (master eq.)

For discrete  $n = 0, 1, \dots$ ,

$$G(z, t) = \sum_{n=0}^{\infty} P_n(t) z^n, \quad G(1, t) = 1, \quad \langle n \rangle = \partial_z G(1, t), \\ \langle n(n-1) \dots (n-k+1) \rangle = \partial_z^k G(1, t).$$

Useful translations:

$$\sum_n n P_n z^n = z \partial_z G, \quad \sum_n (n+1) P_{n+1} z^n = \partial_z G, \\ \sum_n (n-1) P_{n-1} z^n = z^2 \partial_z G.$$

Many birth–death master equations become first-order PDEs for  $G$ .

#### Detailed balance

**Equilibrium condition** (pairwise cancellation) w.r.t.  $P^{\text{eq}}$ :

$$W_{ij} P_j^{\text{eq}} = W_{ji} P_i^{\text{eq}} \quad \forall i, j.$$

It is sufficient for stationarity but stronger than stationarity: a stationary nonequilibrium state may have circulating probability currents. For the canonical distribution,

$$\frac{W_{ij}}{W_{ji}} = \frac{P_i^{\text{eq}}}{P_j^{\text{eq}}} = e^{-\beta(E_i - E_j)}.$$

Kolmogorov cycle criterion (for positive rates): detailed balance holds iff the product of rates around every closed cycle equals the product around the reversed cycle.

#### Metropolis and Metropolis–Hastings

**Algorithm.** To sample a target  $\pi_i$ :

1. From state  $i$ , propose  $j$  using  $q(i \rightarrow j)$ .
2. Accept with

$$A(i \rightarrow j) = \min \left[ 1, \frac{\pi_j q(j \rightarrow i)}{\pi_i q(i \rightarrow j)} \right].$$

Otherwise remain at  $i$ .

3. Discard burn-in; estimate observables from the correlated chain and account for autocorrelation.

For **symmetric proposals** ( $q(i \rightarrow j) = q(j \rightarrow i)$ ) and  $\pi_i \propto e^{-\beta E_i}$ ,

$$A(i \rightarrow j) = \min [1, e^{-\beta(E_j - E_i)}].$$

This construction obeys detailed balance. To converge to a unique target, the chain must be irreducible and aperiodic. Smaller proposals accept often but mix slowly; very large proposals are usually rejected.

## Algebra

#### Trigonometry

**Angle addition formulas:**

$$\sin(a \pm b) = \sin(a) \cos(b) \pm \cos(a) \sin(b)$$

$$\cos(a \pm b) = \cos(a) \cos(b) \mp \sin(a) \sin(b)$$

**Product-to-sum formulas:**

$$\cos(a) \cos(b) = \frac{1}{2} [\cos(a+b) + \cos(a-b)]$$

$$\sin(a) \sin(b) = \frac{1}{2} [\cos(a-b) - \cos(a+b)]$$

$$\cos(a) \sin(b) = \frac{1}{2} [\sin(a+b) + \sin(a-b)]$$

**Using complex exponentials:**

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i} \quad \cos(z) = \frac{e^{iz} + e^{-iz}}{2}$$

$$\cosh(x) = \cos(ix) \quad \sinh(x) = -i \sin(ix)$$

**Double-angle identities:**

$$\cos(2x) = \cos^2(x) - \sin^2(x) \quad \sin(2x) = 2 \sin(x) \cos(x)$$

**Half-angle identities:**

$$\sin\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{2}} \quad \cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\tan\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$$

#### Powers of Trigonometric Functions

$$\cos^2(x) = \frac{1}{2} + \frac{1}{2} \cos(2x) \quad \sin^2(x) = \frac{1}{2} - \frac{1}{2} \cos(2x)$$

$$\cos^3(x) = \frac{3}{4} \cos(x) + \frac{1}{4} \cos(3x) \quad \sin^3(x) = \frac{3}{4} \sin(x) - \frac{1}{4} \sin(3x)$$

#### Functional Derivative

For  $J[f] = \int_a^b \mathcal{L}(x, f(x), f'(x)) \, dx$ , we obtain:

$$\frac{\delta J}{\delta f(x)} = \frac{\partial \mathcal{L}}{\partial f}(x) - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial f'}(x)$$

For  $J[f] = \int \mathcal{L}(x, f(x)) \, dx$ , we obtain:

$$\frac{\delta J}{\delta f(y)} = \frac{\partial \mathcal{L}}{\partial f}(x = y, f(y))$$

#### Integrals

$$\int x e^{ax} \, dx = \frac{e^{ax}}{a} \left( x - \frac{1}{a} \right) \quad \int x^2 e^{ax} \, dx = \frac{e^{ax}}{a} \left( x^2 - \frac{2x}{a} + \frac{2}{a^2} \right)$$

$$\int x \cos(ax) \, dx = \frac{\cos(ax)}{a^2} + \frac{x \sin(ax)}{a}$$

$$\int x \sin(ax) \, dx = \frac{\sin(ax)}{a^2} - \frac{x \cos(ax)}{a}$$

$$\int x^2 \cos(ax) \, dx = \frac{2x}{a^2} \cos(ax) + \left( \frac{x^2}{a} - \frac{2}{a^3} \right) \sin(ax)$$

$$\int x^2 \sin(ax) \, dx = \frac{2x}{a^2} \sin(ax) + \left( \frac{2}{a^3} - \frac{x^2}{a} \right) \cos(ax)$$

$$\int \sin(ax) \cos(bx) \, dx = -\frac{\cos[(a-b)x]}{2(a-b)} - \frac{\cos[(a+b)x]}{2(a+b)}$$

$$\int \cos(ax) \cos(bx) \, dx = \frac{\sin[(a-b)x]}{2(a-b)} + \frac{\sin[(a+b)x]}{2(a+b)}$$

$$\int \sin(ax) \sin(bx) \, dx = \frac{\sin[(a-b)x]}{2(a-b)} - \frac{\sin[(a+b)x]}{2(a+b)}$$

$$\int \sin^n(ax) \cos(ax) \, dx = \frac{\sin^{n+1}(ax)}{(n+1)a}$$

$$\int \cos^n(ax) \cos(ax) \, dx = -\frac{\cos^{n+1}(ax)}{(n+1)a}$$

$$\int e^{ax} \sin(bx) \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin(bx) - b \cos(bx))$$

$$\int e^{ax} \cos(bx) \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos(bx) + b \sin(bx))$$

#### Definite Integrals

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} \, dx = \sqrt{\frac{\pi}{\alpha}} \quad \int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} \, dx = \frac{1}{2} \sqrt{\frac{\pi}{\alpha^3}}$$

$$\int_{-\infty}^{\infty} x^{2n} e^{-\alpha x^2} \, dx = \sqrt{\frac{\pi}{\alpha}} \frac{(2n-1)!!}{(2\alpha)^n} \quad \int_{-\infty}^{\infty} x^{2n+1} e^{-\alpha x^2} \, dx = 0$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2 + i\beta x} \, dx = \sqrt{\frac{\pi}{\alpha}} e^{-\frac{\beta^2}{4\alpha}}$$

$$\int_0^{\infty} e^{-ax} \cos(bx) \, dx = \frac{a}{a^2 + b^2} \quad \int_0^{\infty} e^{-ax} \sin(bx) \, dx = \frac{b}{a^2 + b^2}$$

$$\int_0^{\pi/2} \sin^2(x) \, dx = \int_0^{\pi/2} \cos^2(x) \, dx = \frac{\pi}{4} \quad \int_{-\pi/2}^{\pi/2} \cos(x) \, dx = 2$$

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(x) \, dx = f(b(x)) b'(x) - f(a(x)) a'(x)$$

#### Basic Differential Equations

$$\ddot{Y} + \lambda^2 Y = 0 \Rightarrow Y = A \sin(\lambda x) + B \cos(\lambda x)$$

$$\ddot{Y} - \lambda^2 Y = 0 \Rightarrow Y = A \sinh(\lambda x) + B \cosh(\lambda x) = C e^{\lambda x} + D e^{-\lambda x}$$

$$\ddot{Y} + \lambda = 0 \Rightarrow Y = -\frac{\lambda}{2} x^2 + Ax + B$$

$$r^2 R'' + r R' - \lambda^2 = 0 \Rightarrow R = A r^\lambda + B r^{-\lambda}$$

$$y' + P(x)y = Q(x) \Rightarrow y = e^{-\int P(x) \, dx} \left( \int e^{\int P(x) \, dx} Q(x) \, dx + C \right)$$